

Emily Carver

e.carver@uleth.ca | 403-382-0626 | <https://emcarver.github.io> | she/her

Education

B.Sc. Physics

University of Lethbridge, Lethbridge, AB

September 2022 - May 2026

3.95/4.00 GPA

Research Experience

ASTR 3990: Research-Based Independent Study

September 2025 - December 2025

University of Lethbridge, Lethbridge, AB

Multiwavelength Emission Mechanisms in Black Hole X-ray Binaries

Supervisor: Dr. Alexandra Tetarenko

- Used colour-magnitude diagrams to disentangle components of the complex emission from black hole X-ray binary systems (BHXBs) at ultraviolet, optical, and near-infrared wavelengths.
- Implemented MCMC fitting methods to fit analytical models to observational data of BHXBs in outburst and quiescence.
- Designed user-friendly python code to be used for future efforts in fitting a large dataset of XRB systems and outbursts.

Co-op Work Term: Radio Astronomy Researcher

May 2025 - August 2025

National Research Council Canada, Victoria, BC

Visibility Modelling of Galactic Star-Forming Regions at Millimetre Wavelengths

Supervisors: Dr. Doug Johnstone, Dr. Toby Brown

- Applied visibility modelling techniques to star-forming regions of galaxies observed with the Atacama Large (Sub-)Millimetre Array (ALMA) to demonstrate the applicability of existing visibility generation tools used to study protoplanetary disks. A summary of this work is available on Github¹.
- Identified limitations in the existing code available for generating simulated visibilities, informing future modelling efforts and necessary software development.
- This co-op contributed to an ongoing project in applications of visibility modelling that will be continued by a master's student at the University of Victoria in Fall 2025.

Astrophysics Research Assistant

January 2024 - August 2024

University of Lethbridge, Lethbridge, AB

Investigating the Evolution of Black Hole Jets at Millimetre Wavelengths

Supervisor: Dr. Alexandra Tetarenko

- Investigated time and frequency-based variations in the flux of black hole X-ray binaries GX 339-4 and GRS 1915+105 by analyzing data from ALMA using the python-based CASA software.
- Produced a written report² of my process and findings. A complete set of products is available on Github¹.
- Funded by the highly competitive ULeithbridge Chinook Summer Research Award.
- Results of this project were included in a manuscript I co-authored with international colleagues.

Publications

- [1] P. Gandhi, 35 co-authors including E. Carver, 2026. “*Infrared Synchrotron Emission in the Soft State of GX 339-4 and the Mid-Infrared/X-ray Luminosity Plane of Black Hole X-ray Binaries*”. *MNRAS*, doi: 10.1093/mnras/stag619

Research Presentations

Oral Presentations

Herzberg Astronomy and Astrophysics Science Tea

August 22, 2025

- Presented the results of my co-op project, “Visibility Modelling of Galactic Star-Forming Regions at Millimetre Wavelengths” to an audience of NRC researchers and fellow students at both Dominion Astrophysical Observatory and Dominion Radio Astrophysical Observatory.

¹ <https://github.com/emcarver> ² The report can be viewed here.

*Canadian Conference for Undergraduate Women and Gender Minorities in Physics (CCUW*iP)*

February 1, 2025

- Won second place for my presentation of my research project “*Investigating the Evolution of Black Hole Jets at Millimetre Wavelengths*” at CCUW*iP 2025 hosted by the University of Calgary and University of Alberta

Poster Presentations

CASCA 2026 Annual Meeting

June 9, 2026

- Presented my research project “*Multiwavelength Emission Mechanisms in Black Hole X-ray Binaries*” and accompanying poster to students and researchers at CASCA 2026 hosted by Université de Montréal.

Astrophysics Undergraduate Research Mini-Symposium

September 27, 2024

- Presented my research project “*Investigating the Evolution of Black Hole Jets at Millimetre Wavelengths*” and accompanying poster³ to faculty and students in the Department of Physics and Astronomy, as well as university community members from the wider Faculty of Arts and Science.

University of Lethbridge Summer Research Showcase

August 15, 2024

- Presented my research project and accompanying poster³ to members of the public, fellow undergraduate researchers, and their supervisors.

Professional Experience

Student Success Center Tutor

January 2026 - April 2026

- Developed and hosted a weekly class tutorial for PHYS2130 - *Waves, Sound, and Optics*.
- Built student confidence and knowledge through practice problem walkthroughs and supported work time.

Undergraduate Research Advocate

October 2024 - May 2026

- Developed and led a program to strengthen communication on undergraduate research opportunities, especially those in physics and astronomy.

Department of Physics and Astronomy Student Representative

July 2024 - May 2026

- Chosen by the Physics and Astronomy department head to act as student representative for the 2024/2025 and 2025/2026 academic years.

High School Physics Tutor

December 2024 - January 2025

- Volunteered as a private tutor for a local Physics 30 student, working specifically on preparation for the Alberta Physics Diploma Exam.

Scholarships and Awards

I received the **Dean's Honour List Award** each of my eight completed semesters at ULethbridge. I was also selected as a 2024/2025 **ULethbridge Shining Student**⁴, which recognizes exceptional students at the University of Lethbridge for their academic achievements and community involvement.

Further, I have received numerous merit-based scholarships totalling more than **\$23 500 CAD** over the last four years.

Louise McKinney Post-Secondary Scholarship (\$2500 CAD)	2025
Don and Judi Hall Scholarship in Physics & Mathematics (\$2000 CAD)	2025
Joe Rood Memorial Scholarship in Physics (\$500 CAD) ⁵	2025
Alberta Women in STEM Scholarship (\$2500 CAD)	2025
Jason Lang Scholarship (\$1000 CAD)	2024
Dickinson Memorial Scholarship (\$1850 CAD)	2024
Chinook Summer Research Award (\$6000 CAD)	2024
University of Lethbridge Scholarship (\$1300 CAD)	2023
Jason Lang Scholarship (\$1000 CAD)	2023
University of Lethbridge Achievement Scholarship (\$1500 CAD)	2022

³ A digital copy of the poster is available here. ⁴ My interview can be read here. ⁵ Recipient selected based on outstanding academic achievement in physics and *the greatest potential for research in the physical sciences*.

Board of Governors Admission Scholarship (\$1000 CAD)	2022
Government of Alberta Rutherford Scholarship (\$2500 CAD)	2022
NSERC Undergraduate Student Research Award (\$8730 CAD awarded, declined)	2025
Chinook Summer Research Award (\$6000 CAD awarded, declined)	2025

Outreach

Guest Speaker March 27, 2026

- Invited to speak to 22 students in the grade 5/6 split class at Enchant Elementary School to enrich their learning in the space unit of science class.
- Gave a presentation on modern astronomy and black holes, held an open Q&A, and introduced students to basic sky observing tools.

University of Lethbridge Observatory October 2024 - May 2026

- Team member of an ongoing project to create an observatory at the University of Lethbridge campus to be used for both public outreach and hands-on learning in undergraduate classes.
- Contributed to a project proposal that exceeded initial funding goals.

Astronomy Observing Field Trip Volunteer November 7, 2024; October 18, 2025; February 11, 2026

- Assisted in running the ASTR2020 - Modern Astronomy observing field trip, hosted by Dr. Alex Tetarenko and the Lethbridge Astronomy Society.

University of Lethbridge Open House Volunteer October 26, 2024; October 25, 2025

- Introduced prospective students and their families to the University of Lethbridge, highlighting experiences in physics and astronomy, as well as in undergraduate research.

Shad Canada Lab Assistant July 15, 2024

- Shad Canada is a science and entrepreneurship program for high school students, hosted at Canadian universities over the summer months.
- Answered student questions on math and circuit theory, and guided construction of a simple filter and amplifier circuit to assist in an introductory circuitry lab designed to introduce high school participants to experimental physics at a post-secondary level.